

NOXUS — WEEK 11 & WEEK 12

WEEK 11: OPEN SOURCE, ADVANCED COLLABORATION & COMMUNITY CONTRIBUTION

1. WEEK 11 OVERVIEW

Week Name: Open Source & Advanced Collaboration

Duration: 5–7 Days

Main Objective

After completing the foundation program and major project journey, students now move beyond working only on their own repositories.

Week 11 introduces students to:

- Open-source development.
- Real repository structures.
- Contribution guidelines.
- Issues and pull requests.
- Code reviews.
- Documentation contributions.
- Community collaboration.

The objective is to help students understand:

How developers contribute to projects built by other people.

2. WEEK 11 GOALS

By the end of Week 11, students should:

- Understand the basics of open source.
- Explore real project repositories.
- Understand README and CONTRIBUTING files.
- Learn how to fork a repository.
- Create branches.
- Understand issues.
- Make a meaningful contribution.
- Create a pull request.
- Participate in review.
- Improve documentation or code.

- Record their contribution in Noxus.

3. DAY 1 — INTRODUCTION TO OPEN SOURCE

Training Topic

What Is Open Source?

Topics Covered:

- What open source means.
- Public repositories.
- Open-source licenses.
- Maintainers and contributors.
- Issues.
- Pull requests.
- Code reviews.
- Contribution guidelines.
- Community rules.

Basic Workflow

```
OPEN SOURCE PROJECT
  ↓
EXPLORE REPOSITORY
  ↓
UNDERSTAND DOCUMENTATION
  ↓
FIND CONTRIBUTION
  ↓
MAKE CHANGES
  ↓
CREATE PULL REQUEST
  ↓
REVIEW
  ↓
MERGE / FEEDBACK
```

Activity

Students select a beginner-friendly repository and analyze:

- What the project does.
- Technology stack.
- Repository structure.
- Available issues.
- Contribution rules.

Deliverable

Repository Exploration Report

4. DAY 2 — UNDERSTANDING A REAL REPOSITORY

Training Topic

How to Read Someone Else's Code

Students should learn how to understand an unfamiliar repository.

Repository Exploration Checklist

- ☐ Read README.
- ☐ Check project structure.
- ☐ Read CONTRIBUTING guidelines.
- ☐ Check open issues.
- ☐ Review previous pull requests.
- ☐ Identify technologies used.
- ☐ Run the project locally if possible.
- ☐ Identify beginner-friendly contribution opportunities.

Activity

Students document:

PROJECT NAME

Purpose:

Technology Stack:

Repository Structure:

Interesting Features:

Possible Contribution:

What I Learned:

5. DAY 3 — FIRST CONTRIBUTION

Students make a beginner-friendly contribution.

Possible contributions:

- Fixing documentation.
- Improving README.
- Fixing a typo.
- Adding examples.
- Improving comments.
- Fixing a small bug.
- Adding a small feature.
- Improving project setup instructions.

Contribution Workflow

```
graph TD; A[FORK REPOSITORY] --> B[CLONE REPOSITORY]; B --> C[CREATE BRANCH]; C --> D[MAKE CHANGE]; D --> E[TEST CHANGE]; E --> F[COMMIT]; F --> G[PUSH]; G --> H[CREATE PULL REQUEST];
```

Deliverable

A documented contribution or pull request.

6. DAY 4 — CODE REVIEW & COLLABORATION

Training Topic

Giving and Receiving Technical Feedback

Students learn:

- How to review code.
- How to give constructive feedback.
- How to respond to review comments.
- How to update a pull request.
- How to handle rejected changes.
- How to improve communication.

Peer Review Activity

Students review another student's:

- Code.
- README.
- Documentation.
- Pull request.

Feedback should focus on:

- Clarity.
- Functionality.
- Code quality.
- Documentation.
- Suggestions for improvement.

7. DAY 5 — OPEN SOURCE CONTRIBUTION REVIEW

Students present:

1. Repository selected.
2. Problem or contribution identified.
3. Changes made.
4. Pull request status.
5. Challenges faced.
6. Feedback received.
7. What they learned.

A contribution does not need to be merged during the same week to count as learning progress.

The focus is on making a **genuine and understandable contribution**.

8. WEEK 11 NOXUS CONTRIBUTION PROFILE

Noxus can track:

OPEN SOURCE PROFILE

Repositories Explored: XX

Issues Studied: XX

Issues Created: XX

Contributions Made: XX

Pull Requests Created: XX

Pull Requests Merged: XX

Documentation Contributions: XX

Code Reviews: XX

9. WEEK 11 XP SYSTEM

Activity	XP
Repository exploration	20 XP
Contribution analysis	25 XP
First contribution	50 XP
Pull request created	50 XP
Pull request merged	75 XP
Documentation contribution	30 XP
Code review	25 XP
Contribution presentation	30 XP
Complete Week 11	100 XP

Important Rule

Students should not receive unlimited XP for creating unnecessary or low-quality contributions.

XP should be awarded based on:

MEANINGFUL CONTRIBUTION

+

QUALITY

+

UNDERSTANDING

+

DOCUMENTATION

+

COLLABORATION

10. WEEK 11 COMPLETION CHECKLIST

- ☐ Understand open-source workflow.
 - ☐ Explore a real repository.
 - ☐ Read project documentation.
 - ☐ Identify a contribution opportunity.
 - ☐ Fork and clone a repository.
 - ☐ Create a contribution branch.
 - ☐ Make a meaningful change.
 - ☐ Test the change.
 - ☐ Create or simulate a pull request.
 - ☐ Participate in peer review.
 - ☐ Document the experience.
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WEEK 11 FINAL OUTCOME

By the end of Week 11, students should have:

1. Experience exploring real repositories.
 2. Understanding of open-source workflows.
 3. Experience with forks and branches.
 4. Experience creating a contribution.
 5. Experience with pull requests.
 6. Experience giving or receiving review feedback.
 7. A documented open-source contribution journey.
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WEEK 12: HACKATHON, OPPORTUNITY READINESS & NEXT-LEVEL ROADMAP

1. WEEK 12 OVERVIEW

Week Name: Hackathon & Opportunity Readiness

Duration: 5–7 Days

Main Objective

Week 12 acts as the transition from structured training into the larger Noxus ecosystem.

Students apply everything learned so far in a challenge-based environment while preparing for:

- Hackathons.
- Advanced projects.
- Open source.

- Internships.
- Technical opportunities.
- Internal Noxus teams.
- Mentorship progression.

The main objective is:

Move from guided learning to independent technical growth.

2. WEEK 12 GOALS

By the end of Week 12, students should:

- Participate in a technical challenge or mini hackathon.
 - Work under time constraints.
 - Build and present a solution.
 - Improve technical communication.
 - Evaluate their current skills.
 - Identify skill gaps.
 - Create a next-step learning roadmap.
 - Complete their Opportunity Readiness Profile.
 - Become eligible for advanced Noxus teams and projects.
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3. DAY 1 — HACKATHON INTRODUCTION & PROBLEM RELEASE

Training Topic

From Problem to Prototype

Students receive one or more problem statements.

Possible categories:

- Education.
- Smart Campus.
- AI.
- IoT.
- Automation.
- Sustainability.
- Healthcare Technology.
- Accessibility.
- Productivity.
- Community Solutions.

Hackathon Flow

PROBLEM RELEASE
↓
TEAM FORMATION
↓
IDEATION
↓
SOLUTION DESIGN
↓
TASK DIVISION
↓
BUILD
↓
TEST
↓
DEMO

Deliverable

Selected Problem + Initial Solution Idea

4. DAY 2 — IDEATION & RAPID PLANNING

Teams quickly define:

- Problem.
- Target users.
- Proposed solution.
- Core features.
- Technology stack.
- MVP.
- Team roles.

Rapid Planning Template

PROBLEM:

TARGET USER:

SOLUTION:

CORE FEATURE 1:

CORE FEATURE 2:

CORE FEATURE 3:

TECH STACK:

TEAM ROLES:

MVP:

Important Rule

The team should avoid planning too many features.

A small working solution is better than a large incomplete idea.

5. DAY 3 — RAPID DEVELOPMENT

Teams begin building their MVP.

Priority:

WORKING CORE

↓

BASIC USER FLOW

↓

INTEGRATION

↓

TESTING

↓

POLISH

Mentors can provide guidance but should avoid building the solution for students.

Students should independently:

- Divide tasks.
- Solve blockers.
- Research documentation.
- Test components.
- Integrate their work.

6. DAY 4 — TESTING & FINAL PREPARATION

Teams focus on:

- Core functionality.
- Major bugs.

- Demo preparation.
- Documentation.
- Presentation.

Final Repository Requirements

```
project-name/
├── README.md
├── src/
├── assets/
├── docs/
└── demo/
```

README Must Include

- Problem statement.
- Proposed solution.
- Features.
- Technology stack.
- Installation or usage.
- Team members.
- Contribution summary.
- Future improvements.

7. DAY 5 — HACKATHON DEMO DAY

Each team presents:

1. Problem.
2. Target users.
3. Proposed solution.
4. Key features.
5. Technology stack.
6. Live demo.
7. Technical challenges.
8. Team contributions.
9. Future improvements.

Evaluation Criteria

Area	Weight
Problem Understanding	15%

Area	Weight
Solution	20%
Technical Implementation	20%
Functionality	20%
Innovation	10%
Teamwork	5%
Documentation	5%
Presentation	5%

8. WEEK 12 BADGES

Possible badges:

-  HACKATHON WINNER
-  BEST INNOVATION
-  RAPID BUILDER
-  BEST TEAM PLAYER
-  BEST PROBLEM SOLVER
-  BEST TECHNICAL IMPLEMENTATION
-  BEST PRESENTATION

9. OPPORTUNITY READINESS REVIEW

After the hackathon, every student completes a personal readiness review.

TECHNICAL SKILLS

Programming: XX%

Problem Solving: XX%

Projects: XX%

GitHub: XX%

Domain Knowledge: XX%

Communication: XX%

Teamwork: XX%

Portfolio: XX%

10. SKILL GAP ANALYSIS

Noxus should identify areas where students need improvement.

Example:

CURRENT PROFILE

Strong Areas:

- ✓ HTML / CSS
- ✓ GitHub Basics
- ✓ Team Collaboration

Needs Improvement:

- JavaScript
- Data Structures
- API Integration

NEXT LEARNING GOALS:

1. Complete JavaScript Fundamentals
2. Solve 20 Array Problems
3. Build One API Project

This creates a personalized direction after Week 12.

11. NEXT-LEVEL PATH SELECTION

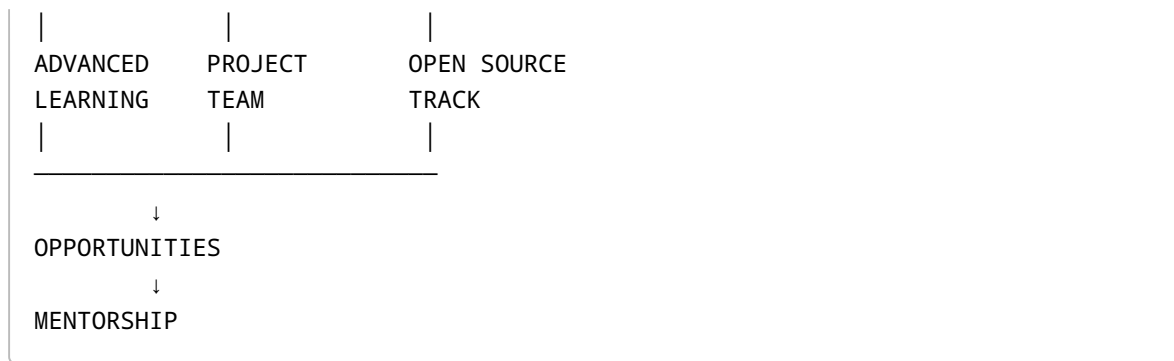
Students can now move into different paths.

WEEK 12 COMPLETE

↓

SELECT NEXT PATH

↓



Possible paths:

PATH 1 — Advanced Learning

For students who need stronger technical foundations.

PATH 2 — Project Builder

For students who want to build more projects.

PATH 3 — Open Source Contributor

For students interested in collaborative development.

PATH 4 — Competitive Programming

For students focused on coding and problem-solving.

PATH 5 — Research & Innovation

For students interested in research, experiments, and technical innovation.

PATH 6 — Technical Team

For students ready to join specialized Noxus teams.

12. OPPORTUNITY READY STATUS

Students can unlock:

STATUS: OPPORTUNITY READY

This should require more than XP.

Suggested Requirements

- Completed foundation journey.

- At least one significant project.
 - Active GitHub profile.
 - Proper project documentation.
 - Resume completed.
 - Portfolio available.
 - Basic technical communication.
 - Participation in collaborative work.
 - Mentor evaluation.
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13. WEEK 12 XP SYSTEM

Activity	XP
Hackathon participation	50 XP
Project ideation	25 XP
Rapid development contribution	50–100 XP
Testing contribution	30 XP
Documentation	25 XP
Final demo	75 XP
Hackathon completion	100 XP
Opportunity readiness review	25 XP
Complete Week 12	150 XP

14. WEEK 12 COMPLETION CHECKLIST

- ☐ Participate in the challenge/hackathon.
 - ☐ Work in a team.
 - ☐ Help define the solution.
 - ☐ Complete assigned tasks.
 - ☐ Contribute to the MVP.
 - ☐ Participate in testing.
 - ☐ Update project documentation.
 - ☐ Present the final solution.
 - ☐ Complete skill assessment.
 - ☐ Identify skill gaps.
 - ☐ Select the next learning path.
 - ☐ Complete Opportunity Readiness Profile.
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WEEK 12 FINAL OUTCOME

By the end of Week 12, every student should have:

1. Experience working under project constraints.
2. Hackathon or challenge experience.
3. Improved problem-solving skills.
4. Experience building a solution rapidly.
5. Improved teamwork and communication.
6. A new documented project.
7. A clear understanding of their strengths.
8. A list of technical skill gaps.
9. A personalized next-step roadmap.
10. Eligibility for advanced Noxus activities.

WEEK 11 & WEEK 12 JOURNEY

WEEK 11

OPEN SOURCE

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Repository Exploration

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Contribution

↓

Pull Request

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Code Review

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Community Contribution

WEEK 12

TECHNICAL CHALLENGE

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Problem Release

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Team Formation

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Rapid Development

↓

Testing

↓

Demo Day

↓

Skill Assessment

↓

Opportunity Readiness



NEXT-LEVEL PATH

FINAL 12-WEEK NOXUS JOURNEY

WEEK 1

Orientation & Foundation



WEEK 2

Programming & Problem Solving



WEEK 3

Technology & Domain Exploration



WEEK 4

Collaboration & Mini Project



WEEK 5

Domain Specialization



WEEK 6

Real-World Development



WEEK 7

Major Project Planning



WEEK 8

Major Project Sprint 1



WEEK 9

Testing & Deployment



WEEK 10

Portfolio & Professional Readiness



WEEK 11

Open Source & Community Contribution



WEEK 12

Hackathon & Opportunity Readiness



ADVANCED NOXUS ECOSYSTEM

THE NOXUS 12-WEEK PHILOSOPHY

**Learn → Practice → Build → Collaborate → Specialize → Contribute → Compete →
Showcase → Grow → Lead**

After Week 12, students should not simply finish a program.

They should enter the **continuous Noxus ecosystem**, where they can choose their next direction:

ADVANCED LEARNING

OR

PROJECT BUILDING

OR

OPEN SOURCE

OR

COMPETITIVE PROGRAMMING

OR

RESEARCH & INNOVATION

OR

NOXUS TECHNICAL TEAMS

↓

REAL-WORLD PROJECTS

↓

INTERNSHIPS / OPPORTUNITIES

↓

MENTORSHIP & LEADERSHIP